Policy and disclosure with endogenous dispersed beliefs

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Abstract

For a declared class of competitive economies with a hidden finite fundamental and heterogeneous private beliefs, we prove the equilibrium and dynamic programming parts of the policy and information problem in E52-44 and treat its disclosure and transparency questions by examples. Household demands, public accounts and the common posterior transition are derived from the primitives. A calibrated probability measure is sufficient for policy, including persistent fundamentals and disclosure channels with zero entries. The resulting discounted Bellman equation characterizes the value and stationary Borel policies that approximate it uniformly. Equally costly private disclosure can strictly outperform the optimized public technology. A separate example identifies the exact posterior threshold at which transparency changes the optimal stabilization tax.

1 The question and the declared economy

The registered problem E52-44 [1] asks, under explicit policy, information and equilibrium-selection restrictions, for optimal or approximating policies in an infinite-horizon competitive economy with dispersed information, dynamic programming on a sufficient state, welfare comparisons between equally costly public and private disclosure, and a characterization of when transparency changes optimal stabilization or the allocation of information across agents. The last of these is treated here only by examples. Its distribution of individual states and beliefs must be generated by equilibrium. The accompanying MA-03 entry, D83-67, requires household optimization, Bayes-consistent beliefs and market clearing; a posterior over current productivity is not sufficient merely by definition. Mankiw and Reis [2, Sections 5.2 and 7.3] provide the policy agenda.

The competitive price is constant, but the fundamental, private posteriors, aggregate demand and their joint distribution are endogenous to policy. No assertion about arbitrary competitive equilibrium correspondences is intended.

There are dates $t = 0, 1, \ldots$, a discount $0 < \beta < 1$, and a nonatomic unit population divided into $g \geq 1$ signal groups of masses $w_i > 0$, with $\sum_i w_i = 1$. Members of a group share its information; each household remains an individual price taker. The fundamental belongs to a finite set $\Theta = \{\theta_1, \ldots, \theta_d\} \subset [1, 2]$, with $d \geq 1$. The finite nonempty control set A specifies taxes $\tau_a \geq 0$ and transition matrices P_a. Fix $0 < c \leq 1/d$ and require

$$P_a(\ell, \ell') \geq c \qquad (a \in A,\ \ell, \ell' \in [d]).$$

The next fundamental has this transition conditional on the current fundamental and control, independently of the remaining past. Iid transitions are allowed; the theorem also covers persistent ones, for example the two-state matrix $\left(\begin{smallmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{smallmatrix}\right)$ with $c = 0.1$.

There are a numeraire and a produced good. Each household receives B numeraire units each date, where $B > 1 + \max_a \tau_a$, and chooses $(c_i^0, q_i) \in [0, B] \times [0, 1]$. There is no storage, saving or individual physical state carried between dates. Its period utility is $c_i^0 + \theta q_i - q_i^2/2 - \kappa Q^2$,

where $\kappa \geq 0$ is fixed and $Q = \sum_i w_i q_i$. A competitive firm has the compact production set $\{(-v, v) : -2 \leq v \leq 2\}$; households own its profits in fixed shares. The government collects the output tax and consumes exactly $G = \tau_a Q$ numeraire units. This public consumption has unit social value; social welfare sums private utilities and this public consumption, counting the external loss κQ^2 once. Individual households cannot influence that aggregate loss or future policy through their own choices.

Information and commitment. Finite nonempty alphabets $Y, Z_1, \ldots, Z_g$ contain a public message and group messages. A prescription k is a probability array $k(a, y, z_1, \ldots, z_g \mid \ell)$ for each ℓ. The feasible set $\mathcal{K}$ is any nonempty compact subset of this finite product of simplexes. Channel entries may be zero. At each date the authority chooses and publicly announces k using the public history, with optional public randomization. It observes the current fundamental and executes this announced array. It may not condition execution on its private past or past private messages. Households observe the realized (a, y) and their own z_i before choosing quantities. Thus the tax itself is an observed signal and is included in the filter. These are explicit commitment and information restrictions on the authority's adapted policy.

The public history contains previous announcements and public outcomes. A household also remembers its group's messages. Neither the current fundamental, realized utility, aggregate orders nor government receipts are separately observed. The public price is observed. Own choices and numeraire consumption are functions of information already held and add no signal. No subsequent unmodeled feedback is supplied. These restrictions prevent a hidden extra learning channel. The firm fills aggregate orders after household choices, and the government administers receipts. Neither observation is shared as household feedback or used as an additional argument of an admissible prescription.

2 Competitive allocation and the sufficient common state

Write $\Delta_c = \{b \in \mathbb{R}^d : b_\ell \geq c,\ \sum_\ell b_\ell = 1\}$. Before current disclosure, b_i denotes group i's posterior over the current fundamental given its entire information then available. The candidate public state is the conditional law π of $(\ell, b_1, \ldots, b_g)$ on $S = [d] \times \Delta_c^g$. Its admissible values form

$$\Pi = \left\{ \pi \in \mathcal{P}(S) : \int (\mathbf{1}_{\{\ell=j\}} - b_{ij}) \varphi(b_i)\, d\pi = 0 \quad \forall i, j,\ \forall \varphi \in C(\Delta_c) \right\}. \tag{1}$$

This is the Bayes calibration identity: conditional on group i's posterior, the fundamental has that distribution. Any $\pi_0 \in \Pi$ is an admissible initial condition. It is implemented by drawing (ℓ, b) from π_0 and showing group i its coordinate b_i. The identity for continuous test functions extends to bounded Borel functions, since continuous functions determine finite measures on a compact metric space [3, Section 2, pp. 4–5]. Hence these initial beliefs are genuine conditional probabilities.

Lemma 1 (Competitive allocation)**.** *For every admissible information history the normalized competitive price is* $(1/2, 1/2)$*. If* $\widehat{b}_i$ *is the posterior after current disclosure, the unique household allocation is*

$$q_i = \mathrm{clip}_{[0,1]} \left(\sum_\ell \theta_\ell \widehat{b}_{i\ell} - 1 - \tau_a \right), \qquad c_i^0 = B - (1 + \tau_a) q_i. \tag{2}$$

Select firm output $v = Q$ *and government consumption* $G = \tau_a Q$*. This is a competitive equilibrium, with realized social welfare*

$$W(\ell, a, \widehat{b}) = B + (\theta_\ell - 1)Q - \frac{1}{2} \sum_i w_i q_i^2 - \kappa Q^2. \tag{3}$$

Proof. At a normalized price (p_0, p_1), the firm's profit is $(p_1 - p_0)v$. If $p_1 > p_0$, its unique supply is 2, which cannot clear household demand $Q \in [0, 1]$. If $p_1 < p_0$, its supply is -2, which also cannot clear it. Thus any equilibrium has $p_0 = p_1 = 1/2$, including exclusion of boundary prices. Profits are then zero and every $v \in [-2, 2]$ is optimal.

At the relative price one, the household budget is $c_i^0 + (1 + \tau_a)q_i \leq B$. Utility is increasing in the numeraire, so it binds. The conditional expected externality is independent of an individual household's choice. Substitution therefore leaves a strictly concave quadratic in q_i whose unconstrained maximizer is the argument of the clipping operation in (2). The bound on B makes every $q_i \in [0, 1]$ affordable. No current choice changes future resources, information or policy; therefore datewise maximization also maximizes the discounted household objective. All utilities are bounded, so the discounted sum exists.

The selected $v = Q$ is profit maximizing. Private, production and public numeraire uses satisfy $\sum_i w_i c_i^0 + Q + G = B$, and produced-good demand equals output. Government spending equals tax revenue, without a deficit. Adding private utilities and G gives (3). The tax transfer cancels exactly once. Strict concavity gives unique household demands; the stated firm and government choices complete the selection. □

Theorem 2 (Endogenous filtering and closure)**.** *The space* Π *is nonempty, compact and metrizable in the weak topology. For every* $\pi_0 \in \Pi$ *and admissible policy, the competitive economy has a Bayes-consistent probability law. The conditional common state remains in* Π*, and the formulas below determine its transition, the endogenous population distribution of allocations and beliefs, and expected welfare. The transition is jointly Feller in* (π, k) *and the reward is bounded and jointly continuous, including channels with zero entries.*

Proof. For any $b \in \Delta_c$, drawing ℓ from b and setting every private prior equal to b gives a member of Π. The compact metric space S is complete and separable. Every probability measure on it is tight, using S itself as the compact set. Billingsley's tightness criterion [3, Section 4, Theorem 4.1, p.10] therefore makes $\mathcal{P}(S)$ weakly compact; the weak topology is metrizable [3, Section 2, p.5]. Every integrand in (1) is continuous, so Π is a closed subset and is compact. Taking $\varphi = 1$ also shows $\pi(\ell = j) = \int b_{ij}\, d\pi \geq c$.

Write $u = (a, y)$ and $z = (z_1, \ldots, z_g)$. For each group define

$$L_i^{k,u,z_i}(\ell) = \sum_{z_{-i}} k(u, z_i, z_{-i} \mid \ell), \qquad D_i = \sum_j b_{ij} L_i^{k,u,z_i}(j).$$

For $D_i > 0$ the posterior and next predictive prior are

$$\widehat{b}_{i\ell} = \frac{b_{i\ell} L_i^{k,u,z_i}(\ell)}{D_i}, \qquad b_i' = \widehat{b}_i P_a. \tag{4}$$

Because $b_{ij} \geq c$, a zero denominator means the likelihood is zero at every fundamental. Every corresponding full profile then has zero probability; assigning it an arbitrary posterior affects no expectation. In all cases used with positive probability, $b_i' \in \Delta_c$.

These formulas condition on the full permitted private history. Initially this follows from calibration and its common-prior implementation. Inductively, given that history and current ℓ, the announced prescription has likelihood $L_i(\ell)$ because execution uses no additional private argument. Bayes' rule gives the first formula; the observed controlled Markov transition gives the second. Own choices, spending and the constant price add no signal. This proves the induction under history-dependent policies and their observed public randomizations.

The public probability of u is

$$h_u(\pi, k) = \int_S \sum_z k(u, z \mid \ell)\, d\pi.$$

For $h_u > 0$, define the next common probability measure $\Phi_u(\pi, k)$ by

$$\int_S f \, d\Phi_u = \frac{1}{h_u} \int_S \sum_z k(u, z \mid \ell) \sum_j P_a(\ell, j) f\big(j, (\widehat{b}_i P_a)_i\big) \, d\pi, \qquad f \in C(S). \tag{5}$$

This is the conditional law after the finite channel draw and next fundamental draw. Repeated draws on the initial probability space with fresh independent random variables construct the economy's law. The state transition is

$$\mathsf{P}(\cdot \mid \pi, k) = \sum_u h_u(\pi, k) \delta_{\Phi_u(\pi,k)}. \tag{6}$$

On branches with $h_u = 0$ choose any fixed member of Π; its weight is zero.

To check calibration after a positive public outcome, fix i, j and a continuous $\varphi(b'_i)$. Multiply its desired calibration identity by h_u. Using (5) gives

$$\int_S \sum_z k(u, z \mid \ell)[P_a(\ell, j) - (\widehat{b}_i P_a)_j] \varphi(\widehat{b}_i P_a) \, d\pi.$$

Sum over z_{-i} and condition on b_i. Calibration replaces the law of ℓ by b_i. For every remaining z_i, the inner coefficient is

$$\sum_\ell b_{i\ell} L_i(\ell) P_a(\ell, j) - D_i (\widehat{b}_i P_a)_j = 0.$$

Thus $\Phi_u \in \Pi$. The lower bound on predictive priors follows directly from $P_a \geq c$, so no support restriction is lost.

For the realized hidden profile and messages, the actual population measure at the decision date and its next predictive belief measure are

$$\mu^{\text{dec}} = \sum_i w_i \delta_{(i, \widehat{b}_i, q_i, c_i^0)}, \qquad \mu' = \sum_i w_i \delta_{(i, \widehat{b}_i P_a)}. \tag{7}$$

Their public conditional laws are generated by the same draws in (5); they are not external inputs. The groups are measurable population intervals, so aggregate quantities are exactly their weighted sums without an independence or continuum sampling assumption. For precision, let H_t be the full hidden history of this constructed process. Its coordinates lie in standard Borel spaces. Regular conditional distributions therefore exist [4, Theorems 2.1.22 and 4.1.17]. Take the conditional history law $\mathfrak{b}_{it} = \mathcal{L}(H_t \mid \mathcal{F}_{it})$ given public history and group i's initial information and accumulated private signals. The actual full population measure is $\mu_t^{\text{full}} = \sum_i w_i \delta_{(i, \mathfrak{b}_{it}, q_i, c_i^0)}$. It is generated by the constructed process; its current-fundamental marginal is (4), and (7) is its decision-belief projection. Full history beliefs may differ even when π agrees. Their remaining components affect only terms independent of a household's choice, since no choice changes future opportunities or the aggregate externality. For the authority, π and the legal prescription determine welfare, joint quantities and the next π law. This is the proved sufficient reduction; it does not claim to reconstruct the whole history-belief measure from π.

The expected selected-equilibrium reward is explicitly

$$r(\pi, k) = \int_S \sum_{u,z} k(u, z \mid \ell) W(\ell, a, \widehat{b}(b, k, u, z)) \, d\pi. \tag{8}$$

It is bounded because $\theta \in [1, 2]$ and $q_i, Q \in [0, 1]$.

It remains to prove continuity through zero channels. For $f \in C(S)$, let $\eta_u(\pi, k)$ be the unnormalized measure in (5). Each term of its test integrand has the form

$$k(u, z \mid \ell) \sum_j P_a(\ell, j) f\big(j, (\widehat{b}_i P_a)_i\big).$$

If its coefficient has a positive limit along a convergent sequence of beliefs and channels, every private likelihood used by that profile has positive limiting mass. Since $b_i \geq c$, all relevant denominators stay away from zero, so the term converges by ordinary continuity. If the coefficient tends to zero, the term's absolute value is at most $\|f\|_\infty k(u, z \mid \ell)$ and tends to zero irrespective of a null posterior convention. The finite sum is therefore continuous on the compact product and uniformly continuous there. Weak convergence of π_n and convergence of k_n imply $\eta_u(\pi_n, k_n) \Rightarrow \eta_u(\pi, k)$ and $h_u(\pi_n, k_n) \to h_u(\pi, k)$. Where the limit mass is positive, normalization gives continuity of Φ_u. Where it is zero, for any $v \in C(\Pi)$ the corresponding transition term is bounded by $\|v\|_\infty h_u(\pi_n, k_n) \to 0$. Summing the public outcomes proves joint Feller continuity of (6). It also makes the transition Borel, since the positive-mass branch maps are continuous and their zero-mass extensions are Borel. Exactly the same weighted argument, using continuous clipped demands and bounded continuous welfare in place of f, proves joint continuity of (8). □

3 Optimal policy on the derived state

All histories and policies in this section obey the information and commitment restrictions above. Let $\mathsf{P}(\, d\pi' \mid \pi, k)$ and $r(\pi, k)$ denote the transition and expected welfare derived in the preceding section. Put $M = \sup_{\pi,k} |r(\pi, k)| < \infty$ and, for $v \in C(\Pi)$, set

$$Q_v(\pi, k) = r(\pi, k) + \beta \int v(\pi')\mathsf{P}(\, d\pi' \mid \pi, k), \qquad Tv(\pi) = \max_{k \in \mathcal{K}} Q_v(\pi, k). \tag{9}$$

Theorem 3 (Value and approximating policies)**.** *The Bellman equation $V = TV$ has a unique continuous solution. It equals the supremum of discounted equilibrium welfare over all admissible public-history policies, including randomized ones. For every $\varepsilon > 0$ there is a deterministic stationary Borel policy with value at least $V - \varepsilon$ at every $\pi \in \Pi$. If $V_0 = 0$ and $V_H = T^H 0$, then*

$$\|V - V_H\|_\infty \leq \frac{M\beta^H}{1 - \beta}. \tag{10}$$

For any continuous candidate v with $\rho = \|Tv - v\|_\infty$, one has $\|V - v\|_\infty \leq \rho/(1 - \beta)$.

Proof. The derived Feller property makes Q_v continuous on the compact product $\Pi \times \mathcal{K}$. Its maximum exists, and $|Tv(\pi) - Tv(\widetilde{\pi})| \leq \sup_k |Q_v(\pi, k) - Q_v(\widetilde{\pi}, k)|$ proves continuity by uniform continuity. Probability normalization gives $\|Tv - Tw\|_\infty \leq \beta\|v - w\|_\infty$. The iterates from zero satisfy $\|V_{h+1} - V_h\|_\infty \leq M\beta^h$. They converge uniformly to a continuous fixed point V, with $\|V\|_\infty \leq M/(1 - \beta)$ and (10). Two fixed points must coincide by the contraction inequality. Applying that inequality to v and V gives $\|v - V\|_\infty \leq \rho + \beta\|v - V\|_\infty$, proving the residual bound.

For any state and prescription, the fixed point implies $V(\pi) \geq r(\pi, k) + \beta\mathbb{E}[V(\pi') \mid \pi, k]$. The equilibrium closure makes this the correct conditional inequality after every public history. Conditioning also on public randomization and iterating gives an upper bound on the first H discounted rewards plus $\beta^H \mathbb{E}V(\pi_H)$. The latter tends uniformly to zero. Thus V bounds the value of every admissible policy.

For $\eta > 0$, compactness and uniform continuity give a finite set $F = \{k_1, \ldots, k_L\} \subset \mathcal{K}$ with $0 \leq V(\pi) - \max_{k \in F} Q_V(\pi, k) \leq \eta$ for every π. Select the first maximizer in this list. Its selection sets are finite intersections of strict or weak comparisons of continuous functions, so the stationary policy f is Borel. Its Bellman deficit $d(\pi) = V(\pi) - Q_V(\pi, f(\pi))$ lies in $[0, \eta]$. Conditional telescoping, followed by the bounded-tail limit, yields

$$V(\pi) - J_f(\pi) = \mathbb{E}_f \sum_{t \geq 0} \beta^t d(\pi_t) \leq \frac{\eta}{1 - \beta}.$$

Taking $\eta = (1 - \beta)\varepsilon$ proves both approximation and equality with the policy supremum. No exact measurable maximizing selector is needed. □

Corollary 4 (Finite approximation and a policy certificate)**.** *Fix $H \geq 1$. A common finite action set $F \subset \mathcal{K}$ can be chosen so that $0 \leq TV_h - \max_{k \in F} Q_{V_h}(\cdot, k) \leq \eta$ for $h = 0, \ldots, H-1$. Let $W_0 = 0$ and $W_h = \max_{k \in F} Q_{W_{h-1}}(\cdot, k)$. Then*

$$0 \leq V_h - W_h \leq \frac{\eta(1-\beta^h)}{1-\beta}.$$

Use the first finite maximizer with the appropriate remaining horizon and then any feasible action after date $H-1$. Its infinite-horizon value J_H satisfies

$$0 \leq V - J_H \leq \frac{\eta(1-\beta^H) + 2M\beta^H}{1-\beta}. \tag{11}$$

More generally, if a finite first-maximizer policy f satisfies $0 \leq Tv - Q_v(\cdot, f(\cdot)) \leq \eta$, then $0 \leq V - J_f \leq (2\rho + \eta)/(1-\beta)$.

Proof. There are finitely many continuous functions Q_{V_h}, so one sufficiently fine finite action net works for all of them. Monotonicity gives $V_h \geq W_h$. If their maximum difference at stage $h-1$ is e_{h-1}, insertion of V_{h-1} between the full and finite maximizations gives $e_h \leq \eta + \beta e_{h-1}$. Summation proves the first bound. Finite first-maximizer policies attain W_H. The same Bellman upper-bound argument as before, together with arbitrarily fine action nets, shows that V_H is the finite-horizon policy supremum. Each discounted tail has absolute value at most $M\beta^H/(1-\beta)$, giving (11). For the last claim, $\|v - Q_v(\cdot, f(\cdot))\|_\infty \leq \rho + \eta$. Telescoping bounds $\|v - J_f\|_\infty$ by $(\rho + \eta)/(1-\beta)$; combine with Theorem 3. □

For a finitely supported initial π, each finite prescription and finite message alphabet leave finite support after filtering. A finite action set and horizon therefore give a finite public decision tree, evaluated by backward recursion using the explicit filter; branches of probability zero contribute zero. Compactness guarantees existence of sufficiently fine action nets but does not supply a computable mesh for an arbitrary compact technology. A numerical error certificate requires a proved uniform continuity modulus and validated evaluations. Sampled states alone do not establish a uniform Bellman residual. The results above are policy and approximation characterizations, without an unproved complexity claim.

4 Public and private disclosure at equal cost

If two feasible technologies satisfy $\mathcal{K}_1 \subset \mathcal{K}_2$ at the same resource cost, every policy under the first is available under the second, with the same equilibrium law. Hence

$$V(\mathcal{K}_2) \geq V(\mathcal{K}_1). \tag{12}$$

In particular, private disclosure weakly dominates public disclosure when its feasible arrays include the public arrays by fixing the private message slots. This is an option comparison, not a claim that forced additional information always improves welfare. If those embeddings are excluded or costs differ, (12) does not establish a ranking.

The following comparison is strict against the optimized public technology. Take two equal-sized groups, $\theta = 1 + \omega$ with $\omega \in \{0, 1\}$, initial probability $1/2$ and every row of P_a equal to $(1/2, 1/2)$. Thus each new fundamental is independent of every past history, and the predictive private priors reset to $1/2$. Set $A = \{0\}$ and $\tau_0 = 0$. Fix binary public and private alphabets. The public technology consists of all public kernels with private messages held constant. The private technology consists of all joint kernels on these same fixed alphabets. Both are compact and have zero resource cost, and the latter includes the former. No initial private information distinguishes the households in the public technology.

Proposition 5 (A strict private advantage)**.** *If* $1/2 < \kappa < 1$*, an admissible private experiment has strictly greater welfare than every public policy. At* $\kappa = 3/4$ *a private experiment yields*

$$V(\mathcal{K}_{\text{priv}}) - V(\mathcal{K}_{\text{pub}}) \geq \frac{11}{10000(1-\beta)} > 0.$$

Proof. Under public disclosure all households share posterior $p = \mathbb{P}(\omega = 1 \mid \text{public information})$ and choose $q_i = Q = p$. Bayes consistency gives $\mathbb{E}p = 1/2$ and $\mathbb{E}[\omega p] = \mathbb{E}p^2$. Thus

$$\mathbb{E}W_{\text{pub}} = B + (1/2 - \kappa)\mathbb{E}p^2 \leq B + \frac{1/2 - \kappa}{4}.$$

The inequality uses $\mathbb{E}p^2 \geq (\mathbb{E}p)^2$ and the negative coefficient. Concealment attains it. This bound holds conditional on every public history under any admissible public policy, because the current fundamental is iid. Consequently the optimized discounted public value is $(B+(1/2-\kappa)/4)/(1-\beta)$. The same upper bound would hold for any finite public alphabet; no union of varying alphabets is used as a compact action set here.

Now keep the public message constant and give each group a binary signal, conditionally independent across groups given ω, with

$$\mathbb{P}(+ \mid \omega = 1) = \tfrac{1}{2} + e, \qquad \mathbb{P}(+ \mid \omega = 0) = \tfrac{1}{2} - e, \qquad 0 < e < \tfrac{1}{2}.$$

Each sign has marginal probability $1/2$. Bayes' rule and household demand give $q_i = 1/2 + e$ after plus and $q_i = 1/2 - e$ after minus. Therefore

$$\mathbb{E}q_i^2 = \tfrac{1}{4} + e^2, \qquad \mathbb{E}[q_i \mid \omega] = \tfrac{1}{2} + 2e^2(2\omega - 1).$$

Conditional independence yields $\text{Cov}(q_1, q_2) = 4e^4$. Since $Q = (q_1 + q_2)/2$,

$$\mathbb{E}[\omega Q] = \tfrac{1}{4} + e^2, \qquad \mathbb{E}Q^2 = \tfrac{1}{4} + \tfrac{1}{2}e^2 + 2e^4.$$

Substitution into the actual competitive welfare (3) gives

$$\mathbb{E}W_{\text{priv}} = B + \tfrac{1}{8} - \tfrac{\kappa}{4} + \frac{1-\kappa}{2}e^2 - 2\kappa e^4.$$

Its gain over the best public experiment is $\frac{1-\kappa}{2}e^2 - 2\kappa e^4$, positive whenever $0 < e^2 < (1-\kappa)/(4\kappa)$. For $1/2 < \kappa < 1$ this condition also ensures $e < 1/2$, as required by the channel probabilities. At $\kappa = 3/4$, $e = 1/10$, the private payoff is $B - 307/5000$, the best public payoff is $B - 1/16$, and their difference is $11/10000$. Repeating this feasible private experiment each date and summing the geometric series proves the discounted bound. It is a lower bound on the optimized private value; optimality of the private witness is unnecessary. □

At fixed tax zero, forced full revelation instead of concealment changes period welfare by $(1/2 - \kappa)/4$. It is strictly harmful when $\kappa > 1/2$, even though enlarging an optional disclosure technology cannot reduce its optimized value. Information affects both useful individual responses and the social loss from correlated aggregate demand.

5 When transparency changes stabilization

Retain the iid fundamental and initial information of the preceding section, but set $\kappa = 1/4$ and allow taxes $\tau \in \{0, 2/5\}$. Use a fixed binary public alphabet and constant private messages. The feasible prescriptions have the form

$$k(a, y \mid \omega) = L(y \mid \omega)\mathbf{1}_{\{a = f(y)\}},$$

where L is any kernel on the fixed alphabet and f is a tax assignment chosen and announced before the current fundamental is used. There are only finitely many f, so this is a compact technology. The control can use public history and the generated public message but cannot separately encode the authority's unannounced private fundamental. Disclosure is again free. A blackout restricts the message to a constant, making its tax constant conditional on history. This restriction is essential to the comparison.

Proposition 6 (The exact tax threshold)**.** *With common posterior p, the optimal tax is zero for $0 < p < 3/5$ and $2/5$ for $p > 3/5$. Both tie at $p = 3/5$ and at $p = 0$. Full public revelation maximizes discounted welfare over the declared disclosure and tax technology. Its gain over blackout is*

$$\frac{41}{400(1-\beta)}.$$

Proof. Household optimization gives $q = Q = (p-\tau)_+$, with no upper clipping since $p \le 1$ and taxes are nonnegative. Conditional expected incremental welfare is

$$H_\tau(p) = p(p-\tau)_+ - \tfrac{3}{4}(p-\tau)_+^2.$$

For tax zero this is $p^2/4$. For tax $2/5$ it is zero when $p \le 2/5$, and when $p > 2/5$ it equals $p^2/4 + p/5 - 3/25$. Comparing the two expressions gives the stated threshold and all ties. The optimized posterior payoff is therefore

$$H(p) = \tfrac{1}{4}p^2 + \max\{0, p/5 - 3/25\}.$$

It is convex on $[0,1]$. For any experiment, $\mathbb{E}p = 1/2$; the convex chord bound gives

$$\mathbb{E}H(p) \le \tfrac{1}{2}H(0) + \tfrac{1}{2}H(1) = \frac{33}{200}.$$

Full revelation attains the bound by setting tax zero in the low state and $2/5$ in the high state. This proves global optimality within the declared discrete tax technology. Under blackout, $p = 1/2$ and tax zero is uniquely optimal, giving $H(1/2) = 1/16$. Thus the period gain is $33/200 - 1/16 = 41/400$.

All physical transition rows are the same and do not depend on tax. The next fundamental is independent of current information and controls; every next predictive prior is $1/2$. Accordingly the same upper bound applies after every public history, including public randomizations. Repeating the maximizing prescription each date attains the discounted optimum. The two values are $(B+33/200)/(1-\beta)$ and $(B+1/16)/(1-\beta)$, proving the claim. Transparency changes stabilization from no tax under blackout to a positive tax in the revealed high state. □

6 Scope

Lemma 1 supplies the selected competitive equilibrium and accounts; Theorem 2 derives its endogenous state law; Theorem 3 and Corollary 4 characterize infinite-horizon optimal value and approximating policies. The disclosure results compare equally costly feasible technologies through the weak inequality (12) and, in examples within the declared class, give a strict private advantage over optimized public disclosure and identify when transparency changes stabilization. Lemma 1 through Corollary 4 cover all finite dimensions, all calibrated initial states, all admissible history policies, $0 < \beta < 1$, strictly positive physical transitions and disclosure arrays with zero cells specified above. They give a solution for this declared class; they do not resolve every macroeconomic model.

The imported probability facts are measure determination by continuous tests, metrizability of weak convergence, and weak compactness on a compact metric space, obtained from the cited

parts of Billingsley, and existence of regular conditional distributions on standard Borel spaces from Durrett. Bayesian filtering, equilibrium, continuity, Bellman verification and examples are proved here.

For the specified competitive class, the proof establishes equilibrium, endogenous filtering and recursion, and discounted policy approximation.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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